Customer Segmentation using RFM Analysis

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1. Introduction:

Customer segmentation is a crucial technique used by businesses to understand customer behavior and improve decision-making. This project focuses on segmenting customers based on their purchasing patterns using the RFM (Recency, Frequency, Monetary) model and visualizing insights through an interactive dashboard.

2. Objective:

The primary objective of this project is to:

- Analyze customer purchasing behavior.
- Segment customers based on RFM metrics.
- Identify high-value and low-engagement customers.
- Provide actionable business insights through visualization.

3. Dataset Description:

The dataset used for this project is Customer Data (sourced from a public dataset).

It contains transactional information such as:

- CustomerID
- Invoice No
- Stock code
- Description
- Quantity
- Invoice date
- Unit Price
- Country

This data enables meaningful segmentation and behavioral analysis.

4. Methodology:

4.1 Data Processing (Python)

The dataset was processed using Python to prepare it for analysis:

- Cleaned and structured transactional data
- Grouped data by CustomerID
- Calculated key metrics:
    - **Recency** → Time since last purchase
    - **Frequency** → Total number of transactions
    - **Monetary** → Total spending

4.2 RFM Scoring

Each customer was assigned scores:

- **R_score** → Based on recency
- **F_score** → Based on purchase frequency
- **M_score** → Based on spending

These scores were used to classify customers into meaningful segments

4.3 Customer Segmentation

Customers were categorized into:

- Champions → High value, frequent buyers
- Loyal Customers → Regular and consistent buyers
- At Risk → Previously active but now inactive
- Others → Low engagement or new customers

5. Dashboard Overview:

An interactive dashboard was created using **Power BI** to visualize insights.

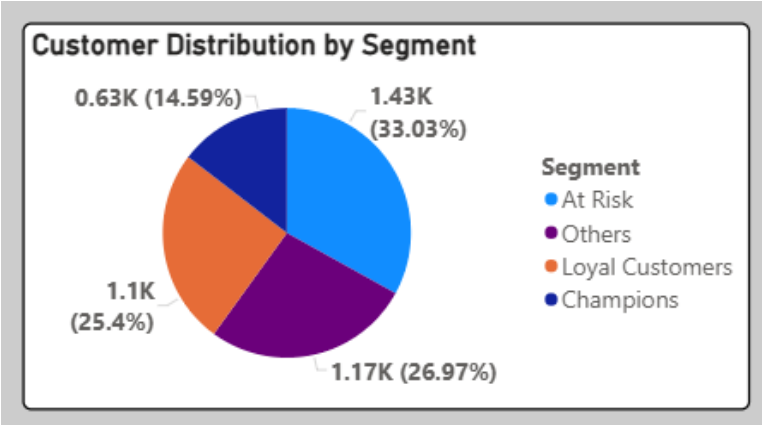
Key Components:

🔹 KPI Cards

- Total Customers: 4.34K
- Total Revenue: 8.91M
- Avg Revenue per Customer: 2.05K

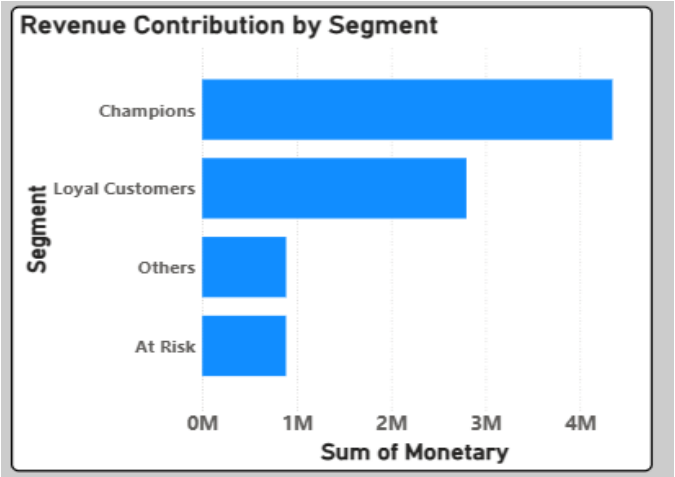
🔹 Customer Distribution (Pie Chart)

Shows how customers are distributed across segments.



🔹 Revenue Contribution (Bar Chart)

Displays revenue generated by each segment.

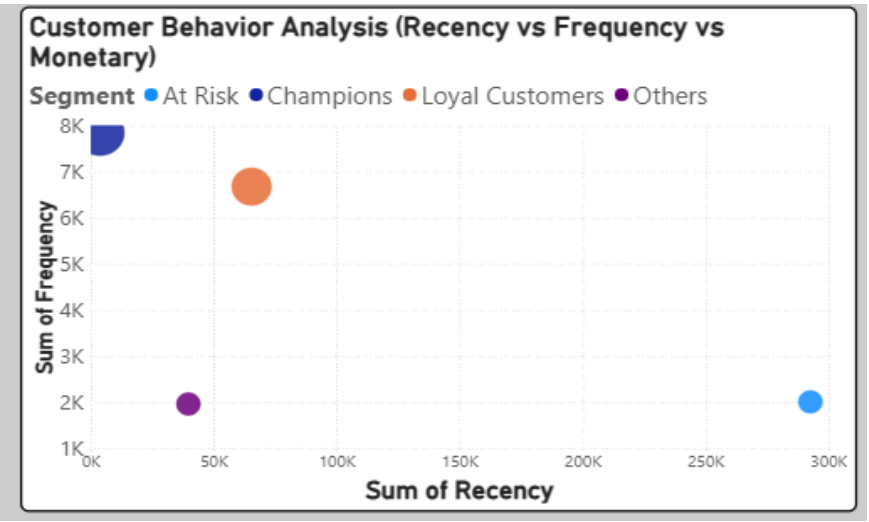


🔹 Customer Behavior Analysis (Scatter Plot)

Analyzes relationship between:

- Recency
- Frequency

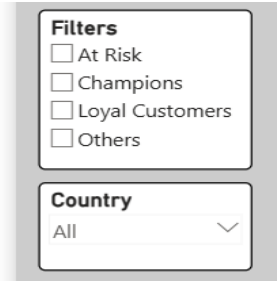
- Monetary



🔹 Filters (Slicers)

- Segment filter
- Country filter

Allows dynamic interaction with the dashboard.



6. Key Insights:

Based on the dashboard:

- Champions segment contributes the highest revenue, making them the most valuable customers
- Loyal customers form a strong base of repeat buyers, ensuring steady revenue
- At Risk customers show low frequency and spending, indicating potential churn
- A significant portion of customers is concentrated in a few segments
- Customers with high recency (inactive) tend to have lower engagement

7. Business Implications:

- Focus marketing efforts on retaining Champions and Loyal Customers.
- Develop re-engagement strategies for At Risk customers.
- Personalize offers based on customer segment.
- Optimize customer retention strategies.

8. Conclusion:

This project demonstrates how RFM analysis can effectively segment customers and provide valuable business insights. By combining Python for data processing and Power BI for visualization, an end-to-end analytical solution was developed.

The dashboard enables businesses to:

- Understand customer behavior
- Identify high-value segments
- Improve decision-making